Urban service allocation: traffic and sensitivity specification

Urban traffic methods

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Purpose and scope. This document specifies a discrete continuous-mass model for an urban signal-control task. It defines lane admission, order-dependent access, service and complete population-time sensitivity. The representative approach has two lanes upstream of a finite left-turn bay and three lanes at the stop line. Numerical calibration, baseline implementation checks and empirical validation are separate from this mathematical definition.

1 Traffic world, action and physical ownership

Retain one isolated four-leg junction, fixed routes and four protected effective-service groups: NS through/right, NS left, EW through/right, EW left. With physical capacity C_ℓ and saturation $s_{\ell m}$,

$$g_p \geq 0, \quad \sum_{p=1}^{4} g_p = 1, \qquad \mu_{\ell m}(g) = \min\{C_\ell, s_{\ell m} g_{p(m)}\}. \tag{1}$$

Mass is continuous vehicle-equivalent quantity; C, s, μ are veh-eq/s. A feasible direction preserves the simplex. Route labels, permissions, topology and source-order skeleton are structural. No phase waveform, integerization, microscopic lane changing or endogenous route choice is introduced. Control uses constant g during the declared control window, then a common candidate-independent clearance allocation.

The representative approach has two parallel feeder lanes and three stop-line lanes:

$$U \longrightarrow \{F_1, F_2\}, \qquad F_1 \longrightarrow \{B, T_1\}, \qquad F_2 \longrightarrow T_2. \tag{2}$$

U is the aggregate upstream road. Within the local zone, F_1 carries left/through traffic and F_2 through/right traffic. The internal bay opening divides F_1 into left-only bay B and through-only continuation T_1; the same cross-section divides F_2 from shared through/right continuation T_2. In the symmetric core $L_{F_1} = L_{F_2}$ and $L_B = L_{T_1} = L_{T_2}$, giving equal total approach path length. All sections own disjoint physical road space and storage.

Left traffic may enter only F_1, then B; right traffic only F_2, then T_2. Through traffic may choose F_1 or F_2 at actual admission from U, but follows its chosen lane thereafter: F_1 to T_1 or F_2 to T_2. There are no cross-feeder lane-change edges and no direct U to B, T_1, T_2 edges. The queue estimate at the choice boundary uses current receiving-feeder mass and earlier actual admissions, as in the allocation rule below; no downstream look-ahead or new driver cost is added. The two physical lanes can therefore differ in accessibility without allowing traffic to escape a lane it already occupies.

V0 uses movement-exclusive full-length paths. V1 uses a shared lane. V2/V3 share the representative bay geometry and permissions: V2 suppresses order-sensitive obstruction by construction, whereas V3 tests it. V4 supplies predeclared held-out variation. The smaller single-feeder case remains a local mechanism diagnostic. Permissions may differ between scenario families, but not between compared models or paired order cases of the same scenario.

Processing order and physical lane order. An unassigned link-end or external-source pool has an ordered processing record. A realized local lane has an actual first-in, first-out sequence. The input record governs allocation inspections; the occupied lane also imposes front blocking. A waiting head in F_1 cannot be passed by a later resident, while an independent F_2 has its own available path. A record cannot be relabeled as unassigned to permit physical overtaking.

2 Propagation and the step clock

Each physical section retains cumulative counts $N^+_{e,k}, N^-_{e,k}$ and occupancy $n_{e,k} = N^+_{e,k} - N^-_{e,k}$ at $t_k = k\Delta t$. Use triangular-FD capacity C_e, free-flow delay τ^f_e, backward delay τ^b_e and storage $K_e = C_e(\tau^f_e + \tau^b_e)$, with $\Delta t \le \min_e(\tau^f_e, \tau^b_e)$. Road capacities, delays and legal saturation rates are finite and positive. An incompatible short section requires a compatible discretization; no fictitious delay, clipping or smoothing is inserted.

With empty prehistory and $\theta = (t - t_h)/\Delta t$,

$$\mathcal{H}^{\pm}_e(t) = (1-\theta)N^{\pm}_{e,h} + \theta N^{\pm}_{e,h+1}, \qquad t_h \le t \le t_{h+1}. \tag{3}$$

The sending count subtracts prior departures from entries early enough to reach the exit. The receiving count starts with storage, adds departures whose space-release wave has reached the entrance, and subtracts all previous entries. Both counts are bounded by capacity times timestep. The inherited one-step limits are

$$A_{e,k} = [\mathcal{H}^+_e(t_{k+1} - \tau^f_e) - N^-_{e,k}]_+, \qquad S_{e,k} = \min(C_e\Delta t, A_{e,k}), \tag{4}$$

$$R_{e,k} = \min\{C_e\Delta t,\ [K_e + \mathcal{H}^-_e(t_{k+1} - \tau^b_e) - N^+_{e,k}]_+\}. \tag{5}$$

The positive part sets negative availability to zero. Free-flow delay controls downstream travel; backward delay controls how soon new room is available upstream. A is ready resident mass for this update; S, R are mass allowances, not instantaneous rates. Queries use already stored boundaries. New inflow cannot traverse its receiving section in the same update.

This is a timestep model. The internal head-service calculation produces endpoint totals; it does not add propagation-history knots. Readiness uses the inherited discrete sending convention, rather than claiming every selected resident was ready at every instant in the step. No event-history refinement is adopted.

These equations and the left-interval objective are inherited from the approved repository M0 transcription, checked against the local LTM and objective code. Their application to the internal section graph is the declared urban composition, not existing implemented M2 behavior.

3 Ledger, eligibility and lane membership

Store positive-mass cohorts, removing exact zero entries without a cutoff. A cohort stores route-table reference, route position, continuous mass, current owner and affine eligibility provenance. Current movement is projected from its route. Internal $U/F_1/F_2/B/T_1/T_2$ crossings preserve route position; the actual junction crossing advances it. Future movement sequences are not duplicated in grid entries.

Only actual admission establishes receiving-lane membership. Unadmitted mass stays with its old owner without a binding future-lane vector. Traffic already in F_1 or F_2 retains that realized lane; its movement and the declared connectivity determine its continuation. Resident traffic cannot be reassigned across feeders or terminal lanes to improve balance.

Retain the exact source order of the unserved record. Before evaluating an operator, coalesce exactly compatible adjacent representation fragments: same route/position, owner, permissions and contiguous affine eligibility with the M1 exact line condition. No tolerance-based mixing or route merging is used. Serialization changes must preserve predictions; they do not define a new physical traffic unit.

The movement grid is a view of the ordered ledger:

$$M_{cm} = \sum_{r:m(r)=m} |I_r \cap B_c|, \tag{6}$$

where I_r, B_c are intervals in cumulative resident mass. Rank is not metres; mixed composition does not reconstruct exact head order. Cells cover the full resident mass-rank range without truncating the ledger. Storage, travel and spillback come from physical histories.

Sending limits transfers, not candidate inspections. Form candidates from ready mass A, cutting an eligibility segment exactly if needed. For an unassigned pool, scan these candidates and spend at most S in total. The candidate list is not restricted to the first S units before access is checked: an earlier blocked movement must not hide later ready traffic with an independently accessible path. For a physical lane, the access rule below forces actual served mass to be a prefix.

This is an explicit adaptation of M1's prefix extraction. After a selection that skips an independently blocked movement, unserved mass keeps its original entry/travel history and relative residual order. Waiting for admission does not restart its travel delay. Every past departure was eligible when served, and eligibility is monotone. Total eligible residual mass therefore remains $\mathcal{H}^+(t_{k+1} - \tau^f) - N_k^-$ even if eligible departures were not a global processing-order prefix. The adaptation is not claimed to exist in current M1 code, nor to establish passing on a represented single lane.

4 Allocation, unassigned residuals and physical access

Let a be the sending owner, r a ready cohort of mass q_r and movement m_r, and ℓ a receiving lane/section. Permission $P_{m\ell} \in \{0,1\}$ and reachability $E_{r\ell} \in \{0,1\}$ encode actual paths/access, not a future green plan.

Initialize residual sending $s = S_{a,k}$, residual receiving $R_\ell^{\text{rem}} = R_{\ell,k}$ and perceived loads $h_\ell = n_{\ell,k}$. Process cohorts in recorded order. Let $\mathcal{B}$ contain movements already left partly unserved in this scan. Each sending owner has its own scan and blocked-movement set; one inbound leg does not block another. If $m_r \in \mathcal{B}$ set $V_r = x_{r\ell} = 0$; otherwise admit the largest feasible total V_r, limited by packet mass, remaining sending and legal reachable receiving. Distribute that total to reduce differences in current lane loads, using a common filling level λ_r:

$$
\begin{aligned}
u_{r\ell} &= P_{m_r\ell} E_{r\ell} R_\ell^{\text{rem}}, \qquad V_r = \min\{q_r, s, \sum_\ell u_{r\ell}\},\\
x_r &= \operatorname*{arg\,min}_{0 \le x_{r\ell} \le u_{r\ell},\ \sum_\ell x_{r\ell} = V_r} \frac{1}{2} \sum_{\ell : P_{m_r\ell}=1} (h_\ell + x_{r\ell})^2,\\
x_{r\ell} &= \min\{u_{r\ell}, [\lambda_r - h_\ell]_+\}, \qquad \sum_\ell x_{r\ell} = V_r.
\end{aligned} \tag{7}
$$

The solver chooses λ_r so the lane additions sum to V_r. Its units are vehicle-equivalent mass: $\lambda_r - h_\ell$ is the mass needed to bring a lane to that load. The positive part excludes lanes already above the level and $u_{r\ell}$ caps each feasible admission. This is a mass-balancing rule, not a stop-line service cap. Illegal lanes have zero addition. Strict convexity selects unique additions; the filling level itself need not be unique when all usable lanes are capped. For $V = 0$, use $x = 0$. Commit only actual entry:

$$
s \leftarrow s - V_r, \quad R_\ell^{\text{rem}} \leftarrow R_\ell^{\text{rem}} - x_{r\ell}, \quad h_\ell \leftarrow h_\ell + x_{r\ell}, \quad q_r^{\text{rem}} = q_r - V_r. \tag{8}
$$

If $q_r^{\text{rem}} > 0$, retain it at its original position and add m_r to $\mathcal{B}$ for this update. Later cohorts of that movement wait; other movements are considered. Reset $\mathcal{B}$ next update. The retained record then puts the older residual first within its movement. There is no waiting-target allocation or old-target rationing law.

h is the selected queue-perception estimate: pre-state local lane mass plus earlier actual admissions. It contains no invented waiting targets and does not subtract forecast signal discharge. Actual next occupancy separately includes that step's exits. Raw queue mass is the first perception model; no service-aware cost or unequal preference is added.

Access in the admitted layouts

For an unassigned pool with independently accessible receiving lanes, $E_{r\ell} = 1$ on those paths. Movement labels alone do not create access. This applies at $U \to \{F_1, F_2\}$ on a packet's legal feeder choices, and to a source feeding independent V0 paths. Once mass occupies F_1 or F_2, it cannot use the other feeder's capacity.

For an occupied single lane, including F_1 at its fork,

$$
E_{r\ell} = \mathbf{1}\{\text{every earlier resident cohort has fully crossed}\} \quad \text{on a legal outgoing edge.} \tag{9}
$$

An older physical residual prevents later physical entry even while the allocator inspects later records. Several actual lanes have separate ledgers/access conditions, not one global FIFO.

Movement-specific blocking governs the unassigned scan; actual lane occupation adds the physical access constraint. At U, a full F_1 prevents left entry while through traffic may still use reachable F_2. At the fork, a through head in F_1 can obstruct a later turner despite an empty bay. A full bay can similarly leave a left

head blocking later through traffic in F_1. Independently resident F_2 traffic may continue toward T_2 if its own sending, receiving and order permit it.

A zero left-turn stop-line rate is not automatically zero lane-entry room. A left turner may join a reachable feeder or bay with room and subsequently contribute to spillback. In this geometry it cannot turn left from T_1 or use F_2, T_2; a full bay has no alternative left-turn stop-line lane. Each constraint refers to its actual physical boundary.

Direct definition checks

The first case below is a local operator diagnostic, not a new primary scenario or a claim that every aggregate road offers independent access. For an independent-access pool with order $[L(0.4), T(0.6)]$, $A = 1$, $S = 0.6$ and exclusive-destination receiving $(0, 0.5)$, left entry is zero and through entry 0.5. Residual order is $[L(0.4), T(0.1)]$. Truncating candidates to the first S mass would expose only 0.2 through mass and give the wrong result for this pool rule.

For actual F_1 take $[T(0.4), L(0.6)]$, $S_{F_1} = 1$, $R_{T_1} = 0.2$, $R_B = 1$. Through entry is 0.2, left entry zero. Reverse the order with the same totals and branch loads $(n_B, n_{T_1}) = (0, 1.8)$. Left uses B first, then through enters 0.2: total 0.8 rather than 0.2. These branch bounds are compatible with $\Delta t = \tau^f = \tau^b = 1$, $C_B = C_{T_1} = 1$, $K_B = K_{T_1} = 2$. If an independent ready 0.3 in F_2 has enough receiving in T_2, it enters in either case. Cross-section totals are then 0.5 and 1.1. These are prepared hand states, not a direct initializer or rollout.

At the U choice boundary, a compatible balancing case has feeder loads $(1.2, 1.0)$, receiving $(0.8, 1.0)$, ready through mass 1 and sending 0.6. Additions $(0.2, 0.4)$ leave perceived loads $(1.4, 1.4)$; the remaining upstream 0.4 is unassigned.

5 Elapsed service and selected receiving rates

A stop-line lane exposes its current ready resident head, with sending quota S_ℓ and cap $c_\ell = \mu_{\ell m}(g)$. The selected coarse receiving convention is

$$\rho_j = R_{j,k}/\Delta t. \tag{10}$$

This is a timestep computational average, not an inferred within-step physical supply curve.

For each output j, collect current heads by inbound leg i. A leg-to-output movement here has one service group:

$$C_{ij} = \sum_{\ell \in \mathcal{L}_{ij}} c_\ell, \quad w_{ij} = g_{p(i,j)}, \quad W_j = \min(\rho_j, \sum_i C_{ij}),$$

$$r_{ij} = \min(C_{ij}, \theta_j w_{ij}), \quad \sum_i r_{ij} = W_j, \tag{11}$$

$$r_\ell = \min(c_\ell, \eta_{ij}), \quad \sum_{\ell \in \mathcal{L}_{ij}} r_\ell = r_{ij}. \tag{12}$$

Remove zero-cap competitors; zero W gives zero rates. Each leg is weighted once; its ready lanes share equally with caps/reuse. Splitting a leg into more software lanes cannot multiply its leg priority.

Hold rates until the earliest head/quota exhaustion or step end, advancing all lanes together. For residual head mass q_ℓ,

$$\delta = \min\left\{\Delta t - \sigma, \min_{\ell: r_\ell > 0} \frac{\min(q_\ell, S_\ell^{\text{rem}})}{r_\ell}\right\}. \tag{13}$$

Transfer $r_\ell \delta$, process tied events together, expose new heads and recompute sharing. Stop when rates vanish or time ends. A zero-rate head blocks its lane. Newly admitted mass is outside this resident service tranche.

A later ready head joins current competition; earlier discharge is not compensated to equalize final totals. Constant receiving removes a separate consumable-pool exhaustion rule, but head clearance and changing competitors remain sequential. Receiving reuse does not transfer one signal group's service to another.

For V1's separate shared left/through service diagnostic, ready left mass q_L precedes sufficient through mass. With nonbinding receiving and physical caps, its clearance time determines the remaining through

service time:

$$y_L = q_L, \qquad y_T = s_T g_T \left(\Delta t - \frac{q_L}{s_L g_L} \right), \quad 0 < \frac{q_L}{s_L g_L} < \Delta t. \tag{14}$$

With unit timestep/saturations, $q_L = 0.2$ and $(g_L, g_T) = (0.4, 0.6)$, through discharge is 0.3. A small transfer from through to left gives through-discharge derivative 0.25 on this branch: faster head removal outweighs the smaller through rate. This is not a complete-TST benefit claim.

6 Paired transfers, histories and feasibility

A one-in/one-out internal edge transfers its ready prefix up to $\min(S_{\text{up}}, R_{\text{down}})$. Forks use allocation/access; the central junction uses elapsed service. An external source adds $D_{o,k}$ to its waiting record before admission, with budget $Q_{o,k} + D_{o,k}$. A sink removes at most sending and its declared allowance.

Each internal entry interface has one sending owner; U has two receiving feeder choices sharing its total sending allowance. Shared receiving at the central signal node uses the simultaneous current-head sharing law, rather than an arbitrary serial order of inbound legs. All nodes read pre-state histories/ledgers. Each section's sending and receiving is spent once at its boundary. New admissions are appended after selecting resident service; today's exits create no immediate receiving credit. The cumulative update is

$$N^+_{e,k+1} = N^+_{e,k} + b_{e,k}, \quad N^-_{e,k+1} = N^-_{e,k} + y_{e,k}, \quad n_{e,k+1} = n_{e,k} + b_{e,k} - y_{e,k}. \tag{15}$$

Internal flow leaves one owner and enters the next with identical mass and route identity. Source admission removes source waiting; sink exit removes road mass. No residual is duplicated as lane stock.

For a receiving lane with total entrance $b > 0$, append children in their actual admission subsequence. A child at entrance rank $[r, r + q']$ gets

$$\left[t_k + \Delta t \, \frac{r}{b}, \; t_k + \Delta t \, \frac{r + q'}{b} \right], \qquad \text{eligibility shifted by } \tau^f_e. \tag{16}$$

For zero inflow create no segment. This is M1-style affine boundary reconstruction applied to the new selection rule, without service-event history knots. Serialize cross-input outgoing records by internal service-interval start, then fixed input-leg index, lane index and source cohort index. This orders macro bookkeeping, not rate grants or a new physical zipper rule. Those roads lead to sinks here; their scalar propagation/total time cannot acquire a later signal-order advantage from that serialization.

The timestep domain gives $S_{e,k} \le n_{e,k}$ and $R_{e,k} \le K_e - n_{e,k}$. Hence

$$0 \le b_{e,k} \le R_{e,k}, \quad 0 \le y_{e,k} \le S_{e,k} \implies 0 \le n_{e,k} + b_{e,k} - y_{e,k} \le K_e. \tag{17}$$

Paired transfers and route-preserving splits imply

$$\sum_o Q_{o,k} + \sum_e n_{e,k} + Z_k = D_{<k}, \tag{18}$$

also route by route. Z_k is cumulative sink exit; $D_{<k}$ all earlier exogenous release.

A finite state has a defined unique result: finite source scan, unique convex allocation vectors, unique capped-sharing vectors and a finite head-event calculation. Each positive service interval exhausts a head/quota or reaches step end. Exact ties are handled together; no mass cutoff or software priority manufactures uniqueness. These are definition-level arguments, not implementation or traffic-fidelity evidence.

7 Sensitivities of the update

Use a fixed structural order and active physical/constraint signature. Differentiate continuous masses, admissions, queues, histories and head times. Labels, permissions, access branches and the source-order skeleton are structural; their switches require branch/one-sided treatment.

For free allocation set $\mathcal{A}$ and upper-capped set $\mathcal{U}$,

$$\lambda = \frac{V - \sum_{\mathcal{U}} u_\ell + \sum_{\mathcal{A}} h_\ell}{|\mathcal{A}|}, \qquad \dot{\lambda} = \frac{\dot{V} - \sum_{\mathcal{U}} \dot{u}_\ell + \sum_{\mathcal{A}} \dot{h}_\ell}{|\mathcal{A}|},$$
$$\dot{x}_\ell = \begin{cases} \dot{\lambda} - \dot{h}_\ell, & \ell \in \mathcal{A}, \\ \dot{u}_\ell, & \ell \in \mathcal{U}, \\ 0, & \text{inactive lower branch.} \end{cases} \tag{19}$$

On stable fully capped branches $\dot{x} = \dot{u}$; stable zero flow has $\dot{x} = 0$. Differentiate V through its active minimum and u through its actual residual receiving/access branch. The scan propagates

$$\dot{h} \leftarrow \dot{h} + \dot{x}, \quad \dot{R}^{\text{rem}} \leftarrow \dot{R}^{\text{rem}} - \dot{x}, \quad \dot{s} \leftarrow \dot{s} - \dot{V}, \quad \dot{q}^{\text{rem}} = \dot{q} - \dot{V}. \tag{20}$$

There is no frozen-waiting-vector derivative. A blocked cohort's zero admission does not erase its retained mass sensitivity. Movement and physical-prefix switches are not smoothed.

For uncapped leg set $\mathcal{F}$ and capped set $\mathcal{C}$,

$$\dot{\theta} = \frac{\dot{W} - \sum_{\mathcal{C}} \dot{C}_{ij} - \theta \sum_{\mathcal{F}} \dot{w}_{ij}}{\sum_{\mathcal{F}} w_{ij}}, \qquad \dot{r}_{ij} = \begin{cases} \dot{C}_{ij}, & i \in \mathcal{C}, \\ w_{ij}\dot{\theta} + \theta \dot{w}_{ij}, & i \in \mathcal{F}. \end{cases} \tag{21}$$

If the uncapped set is empty, differentiate the cap branches directly; a stable zero-supply branch has zero grants. No empty-set division is used. Within-leg sharing uses the unweighted analogue, with the same boundary cases. For an interior event $\delta = q/r$, $\dot{\delta} = \dot{q}/r - q\dot{r}/r^2$; the final interval derivative is minus the preceding duration derivatives. Thus

$$y = \sum_a r_a \delta_a, \qquad \dot{y} = \sum_a (\dot{r}_a \delta_a + r_a \dot{\delta}_a). \tag{22}$$

Head times cannot be frozen merely because outputs are evaluated at timestep boundaries.

Fixed delayed queries have fixed interpolation weights but sensitive history values. For an interior eligibility cut at fixed query t,

$$f = \frac{t-a}{b-a}, \quad \dot{f} = \frac{-\dot{a} - f(\dot{b} - \dot{a})}{b-a}, \quad \dot{q}^{\text{ready}} = f\dot{q} + q\dot{f}. \tag{23}$$

Use whole/empty branches outside $[a, b]$; point segments need their boundary branch. For positive entrance b,

$$\dot{t}_{\text{front}} = \Delta t \left(\frac{\dot{r}}{b} - \frac{r\dot{b}}{b^2} \right), \tag{24}$$

with the analogous tail response. No epsilon repairs zero inflow. Exact-compatible merging must preserve the physical map; a storage-format change is not automatically a physical kink.

8 Complete population time and continued state response

Collect physical stocks, histories, ledgers and sources in X_k, with update $X_{k+1} = F_k(X_k, u_k)$. The inherited cost and its response are

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} \left[\sum_o (Q_{o,k} + D_{o,k}) + \sum_e n_{e,k} \right], \tag{25}$$

$$P_0 = 0, \quad P_{k+1} = F_{k,X} P_k + F_{k,u} \dot{u}_k, \quad \dot{J} = \sum_k (c_{k,X} P_k + c_{k,u} \dot{u}_k). \tag{26}$$

Within control $\dot{u}_k = d$; in fixed common continuation $\dot{u}_k = 0$, while P_k keeps evolving. Empty loading/common preparation is independent of the perturbed candidate; otherwise propagate its response too. No per-vehicle trajectory gradient is constructed.

At fixed state, the immediate cost partial for current service is zero. Fork admission also has no direct green input at fixed pre-state: its feedback comes through earlier queues/history. Stop-line service can respond

directly now. These effects compose into access, spillback, source waiting and eventual sink exit. Internal transfer alone does not remove population. For fixed releases,

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} (D_{\le k} - Z_k), \qquad \dot{J}_{\Delta t} = -\Delta t \sum_{k=0}^{H-1} \dot{Z}_k. \tag{27}$$

Use a common integer horizon qualified to clear the compared action neighborhood, followed by zero occupancy. Finite demand, acyclic routes, positive continuation service and eventual sink availability support qualification; no universal numerical horizon or clearance theorem is asserted. A changing stop counter, residual cutoff or nonclearing case cannot silently supply complete TST. The derivative domain includes clearance and unchanged active signatures; switches need directional branches, not a global smoothness claim.

9 Matched comparisons and native qualification

Models share population, geometry/permissions, action, objective, preparation, continuation and a declared data protocol. Original fine-grained series may pass through each model's own macro converter/estimator. Declare retained variables, memory, update schedule and calibration data. Native subsets are legitimate. Successful macro inference counts against needing exact order; an exact ordered ledger retained inside an estimator remains meso information.

Equal current totals do not establish equality of all lawful histories. State-conditioned replay, observation-updated prediction and forecasting from a common prefix are distinct tasks. State which observations are fixed or regenerated for each derivative claim. A later observation cannot count as a response predicted before it was observed.

Baseline roles remain B0 aggregate, B1 structured movement queues, B2 responsive lane/storage aggregates and B3 this selective model. They are not a universally nested information ladder. Qualify principal instances by soundness/adaptation, task fit and then ease.

B1's working source is Wei et al.'s link queue model, with moving/queued state and native effective green. Its full causal short-region/link/node law must be instantiated; the fixed-speed shortcut is not assumed valid in every spillback regime.

B2's working direction uses Gong et al.'s lane-exit allocation with a sourced storage adaptation, for which Liu–Chang (2011) is a candidate. Exit allocation is not finite-bay entry, and native theorems do not automatically qualify that composition. Exact implementations and adapters still require physical and task-specific checks; the cited native model is not automatically a qualified baseline.

A V2 comparator failure must be understood before attributing a V3 advantage to retained order.

10 Definition checks and external usefulness

Local reference checks cover pool/feeder access, partial admission, finite storage, current-head sharing/reuse, route conservation and the source-counted objective. V0 neutral cases need matched pure service classes and nonbinding or equivalent receiving; one through/right green does not make differently constrained exits identical. In that neutral domain the laws reduce to the same sending/service/receiving minimum. Exact equivalence concerns a matched aggregate backbone; a source-native propagation difference must be qualified and cannot be credited to retained order.

Rotating the symmetric scenario and green groups must rotate results. Equal lane totals do not justify one larger FIFO lane without corresponding order, readiness, access, service and receiving. Hand/reference checks establish definition correctness, not realism.

For R2 qualify one fixed executable mapping. With fixed cycle C^{cyc}, lost intervals totaling $L < C^{\mathrm{cyc}}$, order and offset,

$$G_p = (C^{\mathrm{cyc}} - L) g_p, \qquad s^{\mathrm{eff}}_{\ell m} = s^{\mathrm{phys}}_{\ell m} (C^{\mathrm{cyc}} - L)/C^{\mathrm{cyc}}. \tag{28}$$

Declare program admissibility and the common switch to fixed clearance; do not silently insert minimum green to repair an API. This matches nominal average opportunity, not microscopic startup/access or every phase effect. Official SUMO signal definitions support the interface, not empirical equivalence.

Use a finite master route population and common desired releases. Integer reference vehicles are exogenous population, not rounded internal mass. Fix native lane behavior independently of candidate directions; do not replay B3's private assignments into only one model. Use matched predeclared reference seeds and held-out family IDs, keeping order pairs together.

Reference time is desired-release-to-arrival time: `duration` plus `departDelay`, from official TripInfo. Waiting is already inside duration. Retain uninserted/unfinished members in master-population accounting; clear all demand or report noncompletion, never a completed-trip-only objective.

For each feasible direction, compare predicted sign/ranking and scale with actual richer-evaluator complete-time change under the same action mapping. Disclose observation conditioning. Only after forward/reference and derivative checks pass should matched optimization and held-out usefulness be tested.

11 Scope and evidence limits

The model specifies eligible packet inspection, movement-specific waiting priority, actual-admission balancing, physical lane access, elapsed service, interpolated histories and complete population-time sensitivity. Its parallel approach lanes retain distinct occupancy, storage and order. The bay feeder can obstruct one part of the approach while another lane remains usable, without giving resident traffic an unmodeled bypass.

Feasibility and unique-selection arguments concern the mathematical operators. The analytic examples illustrate local consequences. Implementation correctness, numerical derivative agreement, qualified baselines and external control usefulness require separate evidence.

The source map credits inherited, adapted and newly defined laws. The LTM basis follows the repository transcription of Yperman (2007) and UNsim v0.12.0; the ordered route/timing machinery follows its route-mass specification. Considering independently accessible packets outside the first sending-budget-sized part of an input list is an explicit urban adaptation, as is the physical section geometry. No publication is credited with the whole composed model.